USN	0	2	J	S	T	2	4	U	C	S	0	3	0	Course Code:	24CS350

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester BE Degree Examination

Department of Computer Science and Engineering

DISCRETE MATHEMATICAL STRUCTURES

Duration: 3 Hours | Max. Marks: 100

Instructions:

- Part A is Compulsory
- Part B has internal choice.

PART – A

Q.NO	1	CO	1	CD	L3	PI	1.2.1	MARKS	5+5=10

a) Define logical eqivalence. Prove that for any propositions p,q,r the compound proposition

$[p \rightarrow (q \wedge r)] \Leftrightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$ using truth table.

b) Test the validity of the following argument using rules if inference.

$[(p \rightarrow r) \wedge (q \rightarrow r)] \Rightarrow (p \vee q) \rightarrow r$

Q.NO	2	CO	2	CD	L3	PI	1.2.1	MARKS	10

Let A = {1,2,3,4,6,8,12}. On A, define the partial ordering relation by xRy if and only if x divides y

i) Prove that R is a partial order on A.
ii) Write the Relation matrix.
iii) Draw the Hasse diagram.

Q.NO	3	CO	3	CD	L3	PI	2.6.3	MARKS	5+5

a) For each of the following encoding functions, find the minimum distance between the code words. Indicate the error detecting and error correcting capabilities of each code.

$E : Z_2^2 \rightarrow Z_2^5$ defined by E(00) = 00001, E(01) = 01010, E(10) = 10100, E(11) = 11111

b) The word c = 1010110 is transmitted through a binary symmetric channel. If p = 0.02 is the probability of incorrect receipt of a signal, find the probability that c is received r = 1011111. Determine the error pattern.

Q.NO	4	CO	4	CD	L2	PI	1.7.1	MARKS	10

Define the terms with example:

i) Simple graph

ii) Multigraph
iii) Planar graph.
iv) Bipartite graph
v) Complete Bipartite graph

Q.NO	5	CO	5	CD	L3	PI	1.7.1	MARKS	10

Construct an optimal prefix code for the message ROAD IS GOOD. Decode Message.

PART – B

Q.NO	6	CO	1	CD	L3	PI	1.2.1	MARKS	6+4

a) Test the validity of the following argument using rules of inference.

If i study, i will not fail in the examination.
If i do not watch TV in the evenings, i will study
I failed in the examination.
Therefore i might have watched TV in the evenings

b) For the universe of all integers, Let
P(x) : $x > 0$ **Q(x)** : x is even **R(x)** : x is a perfect square
S(x) : x is divisible by 3 **T(x)** : x is divisible by 7.
Write down the following quantified statements in symbolic form.

i) There exists a positive integer that is even.
ii) Some even integers are divisible by 3.
iii) Every integer is either even or odd.
iv) If x is even and a perfect square, then s is divisible by 3.

OR

Q.NO	7	CO	1	CD	L3	PI	1.2.1	MARKS	6+4

a) Establish the validity of the following argument using rules of inference.

$(\neg p \vee q) \rightarrow r$

$r \rightarrow (s \vee t)$

$\neg s \wedge \neg v$

$\neg u \rightarrow \neg t$

$\therefore p$

b) If $p(x) = x \geq 0$, $q(x)$: $x^2 \geq 0$, $r(x)$: $x^2 - 3x - 4 = 0$, $s(x)$: $x^2 - 3 > 0$, Find the truth values of the following if the universe contains set of real numbers.

i) $\exists x,[p(x) \wedge q(x)]$
ii) $\forall x,[p(x) \rightarrow q(x)]$
iii) $\forall x,[q(x) \rightarrow s(x)]$
iv) $\forall x,[r(x) \vee s(x)]$

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Q.NO	8	CO	2	CD	L3	PI	1.7.1	MARKS	10

Let A={1,2} B={m,n,p} C={3,4,5}. Consider relation $R_1 \subseteq (A \times B)$, $R_2 \subseteq (B \times C)$ and $R_3 \subseteq (B \times C)$ defined by

R_1 = {(1,m), (1,n), (1,p)}

R_2 = {(m,3), (m,4), (n,3)}

R_3 = {(n,3), (n,5), (p,4)}

Determine the following

i) $R_1 \circ (R_2 \cup R_3)$
ii) $R_1 \circ (R_2 \cap R_3)$
iii) $(R_1 \circ R_2) \cup (R_1 \circ R_3)$
iv) $(R_1 \circ R_2) \cap (R_1 \circ R_3)$

OR

Q.NO	9	CO	2	CD	L3	PI	1.71	MARKS	10

Which elements of the POSET ({2,4,5,10,12,20,25} /) are maximal and which are minimal.

Q.NO	10	CO	3	CD	L3	PI	2.6.3	MARKS	10

A encoding function E: $z_2^2 \rightarrow z_2^5$ is given by generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

i) Determine all the code words. What can be said about the error detection capability of this code. What about its error correction capability?

ii) Find the associated parity check matrix H.

iii) Use H to decode the received words 11101, 11011

OR

Q.NO	11	CO	3	CD	L3	PI	2.6.3	MARKS	5+5=10

The generator matrix for an encoding function E: $z_2^3 \rightarrow z_2^6$ is given by

$$G = \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

a) Find the code words assigned to 110 and 010

b) Obtain associated parity-check matrix

c) Hence decode the received words: 110110 and 111101

d) Show that the decoding of 111111 is not possible by using H

Q.NO	12	CO	4	CD	L2	PI	2.6.3	MARKS	10

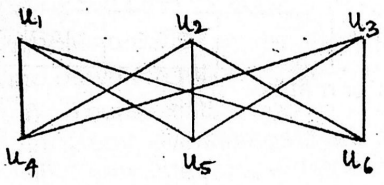
Define the terms with example

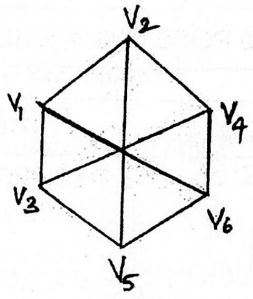
i) Walk
ii) Euler Trail
iii) Euler Circuit
iv) Hamilton Path
v) Hamilton cycle.

OR

Q.NO	13	CO	4	CD	L3	PI	2.6.3	MARKS	10

Define isomorphism. Prove that the two graphs shown in the figure below are isomorphic.





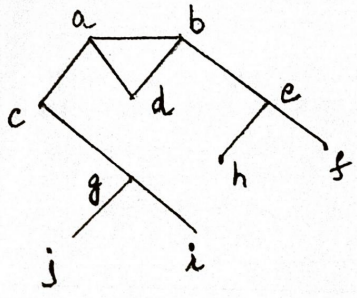
Q.NO	14	CO	5	CD	L3	PI	1.7.1	MARKS	5+5=10

a) Define prefix codes. Construct a binary tree for the given codes. A code for {a,e,n,r,t} is given by a:111, e=0, n= 1100, r:1101, t:10

Decode the sequence

11110111, 111110010, 1100010, 11010110010

b) Find the depth-first spanning tree for the graph shown in the figure if the vertex order is d,a,e,f,g,h,i,b,j,c



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OR									
Q.NO	15	CO	5	CD	L2	PI	1.7.1	MARKS	10

Define the following

i) Spanning tree
ii) Binary tree
iii) Complete binary tree
iv) Weighted tree
v) Expression tree

Course Outcome (CO):	
CO–1	Use the knowledge needed to test the logic of the program.
CO–2	Demonstrate the concept of relations and be able to determine their properties.
CO–3	Explain and apply decode group codes based on the various methods
CO–4	Identify different types of graphs such as Isomorphism, Bipartite, Planar, Hamilton paths and cycles, Euler trails and circuits based on their properties.
CO–5	Explain different types of trees and apply tree traversal techniques like preorder and postorder.

PI	
1.2.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems.
1.7.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.6.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyse
L5	Evaluate
L6	Create

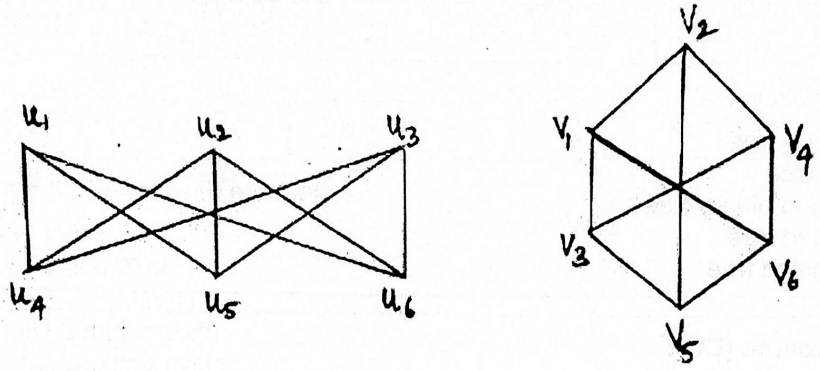


figure 1

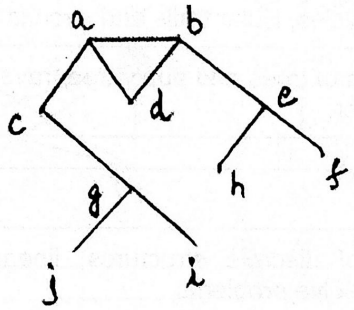


figure 2

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